

Lab Goal : This lab was designed to teach you more about stacks.

Lab Description : Take a postfix expression and solve it. All numbers will be in the range 0 to 9.

Standard infix expression $7 + 8 - 2$

Standard postfix expression $7\ 8\ +\ 2\ -$

Files Needed ::

Postfix.java

Lab13c.java

Sample Data :

2 7 + 1 2 + +
1 2 3 4 + + -
9 3 * 8 / 4 +
19 3 / 2 * 7 9 * + 43 -
35 5 + 2 * 4 / 19 +
2.3 49.2 + 0.27 * 9 2 / +

Sample Output :

2 7 + 1 2 + + = 12.0
1 2 3 4 + + - = -8.0
9 3 * 8 / 4 + = 7.375
19 3 / 2 * 7 9 * + 43 - = 32.666666666666666
35 5 + 2 * 4 / 19 + = 39.0
2.3 49.2 + 0.27 * 9 2 / + = 18.405

algorithm help

```
while there are more values in the original expression
{
    get the next value
    if the value is a number
        push the number onto the stack
    else if the value is an operator
        pop 2 numbers from the stack
        use the operator to evaluate the 2 numbers
        push the resulting number onto the stack
}
return the top value from the stack
```